

AI Encyclopedia-You asked I Pointed

1. Function Description

After the program starts, ask the large model which word on the desktop can represent the object on the desktop. After thinking, the large model will reply with the English word that can represent the object, and the program will control the robotic arm to point to that word.

2. Startup

After entering the Docker container, enter the following commands in the Docker terminal.

```
ros2 launch largemodel largemodel_control.launch.py text_chat_mode:=True
```

Then open a second terminal and input the following command:

```
ros2 run text_chat text_chat
```

Then input "Which English word represents the object in front of you?" in the text_chat terminal and press Enter; if it's the voice version, wake up the voice module and directly say to the voice module "Which English word represents the object in front of you?", then wait for the large model to think and reply, as shown in the figure below:

```
user input: Which English word represents the object in front of you?
okay@, let me think for a moment... \[INFO] [1775208895.908914123] [text_chat_node]: Decision making AI planning:Execute in accordance with the following steps, and only output one action function at a time:
1. Call 'seewhat()' to observe the environment;
2. Reply with what objects are in front of the robot;
3. Call 'seewhat()' to check if there are English words representing the objects;
4. Call 'seewhat()' to obtain the positions of the English words for the objects;
5. Call the 'point_to(x1, y1, x2, y2)' function to point to the English words.
[INFO] [1775208897.684823345] [text_chat_node]: "action": ['seewhat()'], "response": "I'm going to take a look around and see what's in front of me!"
[INFO] [1775208897.333823907] [text_chat_node]: "action": ['seewhat()'], "response": "I see an apple in front of me, and there's the word 'Apple' written nearby!"
[INFO] [1775208912.878859393] [text_chat_node]: "action": ['seewhat()'], "response": "I found the English word 'Apple' that represents the object in front of me!"
[INFO] [1775208921.635434413] [text_chat_node]: "action": ['point_to(270, 80, 560, 230)'], "response": "The word 'Apple' is located at the top right of the image, and I'm going to point to it now!"
[INFO] [1775208950.959053889] [text_chat_node]: "action": ['finishtask()'], "response": "I've successfully identified and pointed to the English word 'Apple' that represents the object in front of me!"
```

Then the robotic arm will point to the word Apple:



3. Task Planning

1. Call `seewhat()` to observe the environment;
2. Reply with what objects are in front of the robot;
3. Call `seewhat()` to check if there are English words representing the objects;
4. Call `seewhat()` to obtain the positions of the English words for the objects;
5. Call the `point_to(x1, y1, x2, y2)` function to point to the English words.